Chapter 9. The Origins of Graph Theory

Dots and Lines

Contents

9.1 The notion of graphs. Eulerian trails **2**

9.2 Hamiltonian cycles **3**

9.3 Directed graphs **4**

9.4 The notion of isomorphisms **5**

9.5 Selected exercises **5**

Euler (1736): can one walk through Königsberg, returning to the start, crossing each of its seven bridges exactly once?

9.1 The notion of graphs. Eulerian trails

Motivation. Only the pattern of connections matters, and the answer depends only on degree parities.

A **graph** has finitely many **vertices** and **edges**, each joining two vertices; the **degree** of a vertex is the number of edge-ends at it. Multiple edges and **loops** are allowed; a graph with neither is **simple**.

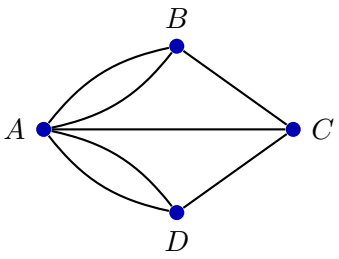


Figure. *Königsberg as a graph: degrees* $5, 3, 3, 3$ *are odd, so no closed Eulerian trail (Theorem 9.2).*

A **trail** is a sequence of distinct edges $e_1 \cdots e_k$ with the end of e_i the start of e_{i+1}; a **walk** may repeat edges. A trail is **closed** if it ends where it starts, **Eulerian** if it uses all edges, and a **path** if it repeats no vertex.

Definition 9.1 (Connected)

G is **connected** if any two vertices are joined by a path.

A **subgraph** H has vertices and edges among those of G; it is **induced** if it contains all edges of G between its vertices. **Connected components** are the maximal connected subgraphs.

Theorem 9.2 (Euler)

A connected graph has a closed Eulerian trail iff all its degrees are even.

Proof.

Step 1 (only if). Along a closed Eulerian trail from S, each visit to $A \neq S$ uses two new edges, so $\deg A = 2a$ (a visits); $\deg S = 2t + 2$ (first and last edge, t intermediate visits).

Step 2 (a first closed trail). Let all degrees be even. From S walk along unused edges. On entering a vertex $\neq S$ an unused edge leaves it (even degree), so we can only stop at S; by finiteness we do, giving a closed trail C_1.

Step 3 (a vertex with unused edges). If $C_1 \neq G$, some vertex V of C_1 has an edge not in C_1: otherwise take $A \notin C_1$; a path from A to C_1 first meets C_1 at some V through an edge not in C_1, a contradiction.

Step 4 (splicing). Removing C_1 keeps degrees even, so by Step 2 there is a closed trail C_2 from V in the remaining edges. C_1 to V, then C_2, then the rest of C_1 is a longer closed trail. Repeat; finiteness gives a closed Eulerian trail. ∎

Corollary 9.3

A connected graph G has an Eulerian trail from S to a different vertex T iff S and T have odd degree and all other degrees are even.

Proof. Add an edge ST to get H. Eulerian S–T trails of G correspond to closed Eulerian trails of H, and H has the degrees of G with $\deg S, \deg T$ raised by 1. Apply Theorem 9.2 to H. ■

Theorem 9.4 (Handshake lemma)

In a graph without loops, the number of vertices of odd degree is even.

Proof. Each edge adds 1 to the degree of each endpoint, so $d_1 + \cdots + d_n = 2e$ is even; hence the number of odd summands is even. ■

9.2 Hamiltonian cycles

Motivation. Visiting every vertex once is NP-complete, but enough edges (Dirac) force it.

A **cycle** is a closed trail repeating no vertex except start = end. A **Hamiltonian cycle** (**path**) is a cycle (path) through all vertices, e.g. a round-table seating where neighbours know each other. Deciding whether one exists is **NP-complete**: a cycle is easy to verify, but no polynomial algorithm is known to certify that none exists (book, Chapter 20).

Theorem 9.5 (Dirac)

Let $n \geq 3$ and let G be a simple graph on n vertices with all degrees at least $n/2$. Then G has a Hamiltonian cycle.

Proof.

Step 1 (connected). If G were disconnected, some component would have $< n/2$ vertices, hence vertices of degree $< n/2$.

Step 2 (a saturated counterexample). Suppose G has no Hamiltonian cycle; add edges while none is created. The result G' has degrees $\geq n/2$, no Hamiltonian cycle, and adding any missing edge creates one; $G' \neq K_n$ since K_n is Hamiltonian.

Step 3 (a Hamiltonian path). For non-adjacent x, y in G', adding xy creates a Hamiltonian cycle through xy, so G' has a Hamiltonian path $x = z_1, \ldots, z_n = y$.

Step 4 (pigeon-hole). Let $A = \{i \in \{2, \ldots, n-1\} \mid xz_i \in E\}$, $B = \{i \in \{2, \ldots, n-1\} \mid z_{i-1}y \in E\}$. As $x \not\sim y$, $|A| = \deg x \geq n/2$, $|B| = \deg y \geq n/2$, so in the $(n-2)$-set $\{2, \ldots, n-1\}$, $|A| + |B| \geq n > n-2$ gives some $i \in A \cap B$.

Step 5. Then $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$ is a Hamiltonian cycle in G', a contradiction. ■

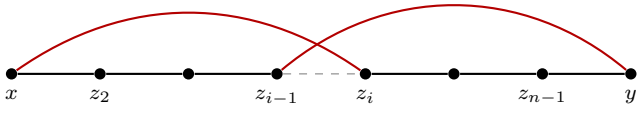


Figure. *Dirac: with $x \sim z_i$ and $z_{i-1} \sim y$ (red), dropping $z_{i-1}z_i$ gives a Hamiltonian cycle.*

9.3 Directed graphs

Motivation. For digraphs Euler's criterion becomes "in-degree = out-degree"; tournaments always have Hamiltonian paths.

In a **digraph** each edge has a direction that trails must follow. G is **strongly connected** if every a has a directed path to every b, and **balanced** if in-degree = out-degree at every vertex. A vertex is **isolated** if both degrees are 0.

Theorem 9.6

A digraph without isolated vertices has a closed Eulerian trail iff it is balanced and strongly connected.

Proof.

Step 1 (necessity). A closed Eulerian trail leaves each vertex as often as it enters it (balanced), and passes through every vertex (strongly connected).

Step 2 (sufficiency) [Supplement]. As in Theorem 9.2: from S walk along unused edges; by balance we can only stop at S, giving a closed trail C_1. If C_1 misses an edge, some V on C_1 has an unused edge: otherwise take $A \notin C_1$ (no isolated vertices) and a directed path from C_1 to A; its first edge leaving C_1 starts at such a V. Removing C_1 keeps balance, so there is a closed trail C_2 from V; splice it into C_1 and repeat. ■

A simple graph is **complete** (K_n) if all $\binom{n}{2}$ pairs are adjacent. An orientation of K_n is a **tournament** (ij an edge if i beat j); there are $2^{\binom{n}{2}}$ on n labelled vertices.

Theorem 9.7

Every tournament has a Hamiltonian path.

Proof. Induction on n; $n \le 2$ is clear. By induction $T - V$ has a Hamiltonian path $h_1 \cdots h_{n-1}$. If $h_i V, V h_{i+1} \in E$ for some i, insert V between them. Otherwise there is k with $V h_j \in E$ for $j \le k$ and $h_j V \in E$ for $j > k$: if $k \ge 1$ put V first, if $k = 0$ put V last. ■

Theorem 9.8 (Camion)

A tournament has a Hamiltonian cycle iff it is strongly connected.

Proof.

Step 1 (only if). A Hamiltonian cycle joins any two vertices by a directed path.

Step 2 (a cycle exists). Let T be strongly connected. With no cycle, $xy, yz \in E$ force $xz \in E$ (else zx closes a 3-cycle), so T is transitive, all edges point right in some ordering, and T is not strongly connected ($n \ge 2$).

Step 3 (a longest cycle). Let $C = y_1 \cdots y_k$ be a longest cycle missing a vertex. Some edge leaves C, say $y_1 x$, $x \notin C$. If $x y_2 \in E$, $y_1 x y_2 \cdots y_k$ is longer; so $y_2 x \in E$, and inductively $y_i x \in E$ for all i.

Step 4. Let $Z = \{z \notin C \mid y_1 z \in E\}$; by Step 3, $y_i z \in E$ for all i, $z \in Z$. Strong connectivity gives $zt \in E$ with $z \in Z$, $t \notin Z$; then $t \notin C$ and $t y_1 \in E$. So $y_k \to z \to t \to y_1 \to \cdots \to y_k$ is a longer cycle, a contradiction. ■

9.4 The notion of isomorphisms

Motivation. Isomorphism makes "same up to relabelling the vertices" precise.

Two graphs on the same labelled vertex set are **identical** if every pair is joined by the same number of edges in both.

Definition 9.9 (Isomorphism)

Graphs G, H are **isomorphic** if there is a bijection f from the vertices of G onto those of H such that X, Y and $f(X), f(Y)$ are joined by the same number of edges, for all X, Y. Such f is an **isomorphism**.

Example 9.10

For the two 4-vertex graphs in the book's Figure 9.8, $f(A) = H$, $f(B) = E$, $f(C) = F$, $f(D) = G$ is an isomorphism; the graphs are not identical (the first contains AB).

Isomorphisms preserve degrees and numbers of cycles and paths of each length, so different degree multisets prove non-isomorphism (equal ones do not prove isomorphism). No efficient general test is known, nor whether the problem is NP-complete; Babai's quasi-polynomial algorithm is a major advance.

9.5 Selected exercises

Motivation. Degree counting, Eulerian trails and longest paths force substructures.

Exercise 9.3 (A 4-cycle from many edges)

Let G be a simple graph on 10 vertices with 28 edges. Prove that G contains a cycle of length 4.

Solution.

Step 1 (cherries). The number of pairs $(v, \{x, y\})$ with $x \neq y$ neighbours of v is $\sum_v \binom{d_v}{2}$.

Step 2 (if there is no 4-cycle). If $v \neq w$ had two common neighbours $x \neq y$, then $v\,x\,w\,y$ would be a 4-cycle. So each $\{x, y\}$ occurs at most once, and $\sum_v \binom{d_v}{2} \leq \binom{10}{2} = 45$.

Step 3 (the degrees force more). $\sum_v d_v = 56$ and by Cauchy–Schwarz $\sum_v d_v^2 \geq \frac{1}{10}\big(\sum_v d_v\big)^2 = 313.6$, so

$$\sum_v \binom{d_v}{2} = \frac{1}{2}\Big(\sum_v d_v^2 - \sum_v d_v\Big) \geq \frac{313.6 - 56}{2} = 128.8 > 45,$$

contradicting Step 2. ∎

Exercise 9.6 (Two orange and two blue edges at each vertex)

Let G be a simple graph in which every vertex has degree 4. Prove that the edges of G can be coloured orange and blue so that every vertex is incident to two orange and two blue edges.

Solution. Colour each component H separately. By Theorem 9.2, H has a closed Eulerian trail $v_0, e_1, v_1, \ldots, e_m, v_m = v_0$.

Step 1 (the colouring). $2m = 4|V(H)|$, so m is even. Colour e_i orange for odd i, blue for even i; put $e_{m+1} = e_1$.

Step 2 (edges at a vertex). Each edge at v has exactly one end at v and occurs once in the trail, so the four edges at v split into the two pairs $\{e_i, e_{i+1}\}$ with $v_i = v$ (for $i = m$ the trail closes via e_1).

Step 3 (conclusion). i and $i+1$ have different parity (also for $\{e_m, e_1\}$, as m is even), so each pair has one edge of each colour. ■

Exercise 9.23 (High minimum degree forces connectivity)

(a) Prove that a simple graph on ten vertices in which every vertex has degree at least five is connected.
(b) Generalize to simple graphs on n vertices.

Solution. (b): a simple graph on n vertices with all degrees $\geq \frac{n-1}{2}$ is connected; $n = 10$ gives (a).

Step 1. For non-adjacent $u \neq v$, $N(u), N(v) \subseteq V \setminus \{u, v\}$ and

$$|N(u)| + |N(v)| \geq n - 1 > n - 2,$$

so some w is adjacent to both, and $u\,w\,v$ is a path.

Step 2. Hence any two vertices are joined by a path of length ≤ 2.

Sharp: for even n, two disjoint copies of $K_{n/2}$ have degrees $\frac{n}{2} - 1$. ■

Exercise 9.46 (A long cycle in a regular graph)

Let $k \geq 2$ and let G be a simple graph in which every vertex has degree k. Prove that G contains a cycle of length at least $k + 1$.

Solution.

Step 1 (a longest path). Let $P = v_0 v_1 \cdots v_m$ be a path with the most vertices.

Step 2 (neighbours of an end). Every neighbour of v_0 lies on P (else P extends). The k distinct neighbours lie among $v_1, \ldots, v_m$, so the largest j with $v_j \sim v_0$ has $j \geq k \geq 2$.

Step 3 (the cycle). $v_0 v_1 \cdots v_j v_0$ is a cycle of length $j + 1 \geq k + 1$.

Only $\min \deg \geq k$ is used; K_{k+1} attains the bound. ■